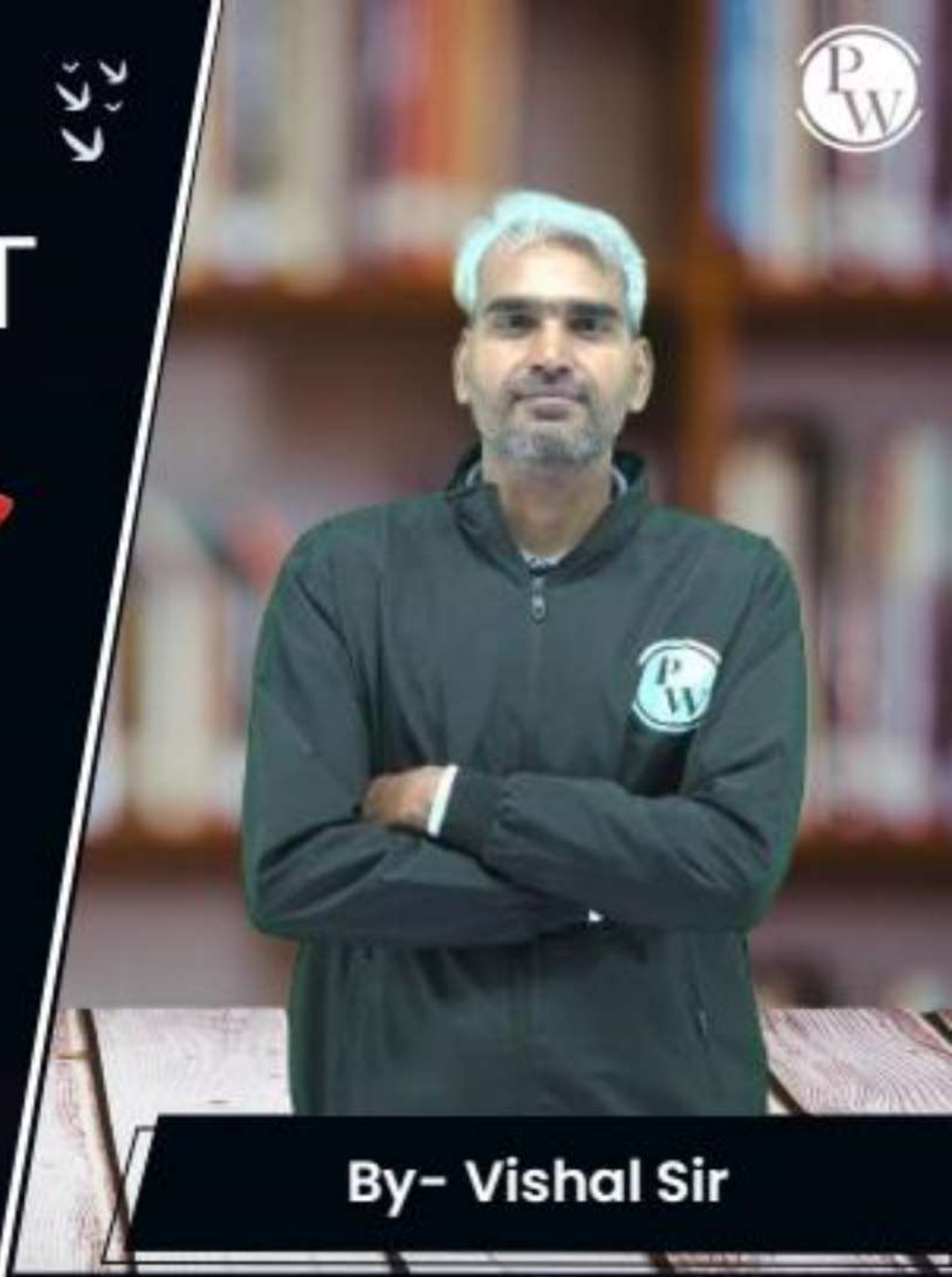
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



Topic Functional dependency

Topic Types of functional dependency

Topic Properties of functional dependency

Slide

Topics to be Covered



- **Topic** Different types of keys in RDBMS
- **Topic** Closure of an attribute set
- **Topic** Super key
- **Topic** Candidate key

Slide

H.W:

$R(A\ B\ C\ D)$

How many non-trivial FDs are Possible:

$$\begin{cases} ABC \to D \\ ABD \to C \\ ACD \to B \\ BCD \to A \end{cases}$$

$A \to B$	$C \to A$	$A \to BC$	$C \to AB$	$A \to BCD$	$AB \to C$	$AB \to D$	$BC \to A$	$BC \to D$
$A \to C$	$C \to B$	$A \to BD$	$C \to AD$	$B \to ACD$				
$A \to D$	$C \to D$	$A \to CD$	$C \to BD$	$C \to ABD$	$AC \to B$	$AC \to D$	$BD \to A$	$BD \to C$
$B \to A$	$D \to A$	$B \to AC$	$D \to AB$	$D \to ABC$				
$B \to C$	$D \to B$	$B \to AD$	$D \to AC$		$AD \to B$	$AD \to C$	$CD \to A$	$CD \to B$
$B \to D$	$D \to C$	$B \to CD$	$D \to BC$					
					$AB \to CD$		$BC \to AD$	
					$AC \to BD$		$BD \to AC$	
					$AD \to BC$		$CD \to AB$	

★ R(A,B,C,D) ✓

28 + 18 + 4 = 50

X ⟶ Y

X: 1 element

Y: 1 or 2 or 3

$X \cap Y = \phi$

* ① One element in LHS

(or)

② Two elements in LHS

(or)

③ Three elements in LHS.

X ⟶ Y

Size 1 (or) Size 2 (or) Size 3

$^4C_1$ and * $\{^3C_1 + {}^3C_2 + {}^3C_3\} = 4 * 7$

$3 + 3 + 1 = 28$

X ⟶ Y (X: two elements)

Size 1 (or) Size 2

$^4C_2 * \{^2C_1 + {}^2C_2\} = 6 * 3 = 18$

(2 and 1)

X ⟶ Y (X: three elements)

Size can be '1' only

$^4C_3 * {}^1C_1 = 4 * 1 = 4$

Q: let $R(A, B, C, D, E)$ be the relational schema. How many FDs of type $X \longrightarrow Y$ are possible from the attributes of relation R, s.t. X & Y are the attribute sets from the attributes of relation R and $X \cap Y = \phi$

In $X \to Y$ if $X \cap Y = \emptyset$ then
X may be of size 1, 2, 3 or 4

180 Ans

① One element in LHS $X \longrightarrow Y$ (1 or 2 or 3 or 4)

$${}^5C_1 * ({}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4) = 5 * (4+6+4+1) = 75$$

② Two elements in LHS $X \longrightarrow Y$ (1 or 2 or 3)

$${}^5C_2 * \{{}^3C_1 + {}^3C_2 + {}^3C_3\} = 10 * \{3+3+1\} = 70$$

③ Three elements in LHS $X \longrightarrow Y$ (1 or 2)

$${}^5C_3 * \{{}^2C_1 + {}^2C_2\} = 10 * \{2+1\} = 30$$

④ Four elements in LHS $X \longrightarrow Y$ (1)

$${}^5C_4 * {}^1C_1 = 5$$



Topic : Different types of keys

- Key is a very important Concept w.r.t. relational database
- In a relation no two tuples (record) can be exactly same.

Key: Key is a set of attributes that can differentiate each record of the relational table uniquely

All the values in the table corresponding to attributes {attributes of key} will be unique

* Their are various types of keys in RDBMS

1. Candidate key
2. Primary key
3. Alternate key
4. Super key
5. Foreign key ← It will be discussed later in the Course

eg ① Consider the Following Relational schema

Student (Sid, Sname, Branch)

let following FDs exists in the Relⁿ

Sid → Sname

Sid → Branch

Sid	Sname	Branch
S_1	A	CS
S_2	A	CS
S_3	B	CS
S_4	C	IT
S_5	B	IT

eg ② Consider the Following relational schema

Enroll (Sid, Cid, Instructor_id (I_id))

let following FDs exists in the relation

Sid, Cid → I_id

Cid → I_id

Sid	Cid	I_id
S_1	C_1	001
S_1	C_2	001
S_2	C_1	001
S_2	C_3	002

Note:- If values are always unique in all tuples corresponding to the attributes of a set of attributes then that set of attribute is called a key of that relation



Topic : Candidate key

Minimal set is a set from which nothing can be removed without destroying its property

The minimal set of attributes that can differentiate each record of the table is called Candidate key

eg ① Consider the Following Relational schema

Student (Sid, Sname, Branch)

let following FDs exists in the Relⁿ

Sid → Sname

Sid → Branch

Sid	Sname	Branch
S_1	A	CS
S_2	A	CS
S_3	B	CS
S_4	C	IT
S_5	B	IT

In eg ①

(Sid, Sname) together is a key of the relation, but it is not a minimal key.

Because even if we remove the attribut "Sname" the set then "Sid" alone can differentiate each record of the table uniquely {i.e. it does not loose its property of being a key}

* (Sid) is a candidate key of the relation w.r.t. eg. (1)

* In eg. ② (Sid, Cid) together is a key of the relation

  - If we remove "Cid" from the set then Sid alone can not differentiate each record of the table {i.e., Sid alone is not a key}
  - If we remove "Sid" from the set then "Cid" alone can not differeniate each record of the table {i.e, Cid alone is not a key}

eg ② Consider the following relational schema

Enroll (Sid, Cid, Instructor_id)
(I_id)

let following FDs exists in the relation

Sid, Cid ⟶ I_id
Cid ⟶ I_id ,

Sid	Cid	I_id
S_1	C_1	001
S_1	C_2	001
S_2	C_1	001
S_2	C_3	002

* In eg ② (Sid, Cid) together is a key, and no attribute can be removed from the set (Sid, Cid) without destroying its property of being a key.

∴ (Sid, Cid) is a minimal set w.r.t. key.

Hence, (Sid, Cid) together is a Candidate key.

* **Note:-** A set of attributes w.r.t. Candidate key can never contain any redundant attribute



Topic : Candidate key

(1) If Candidate key is formed of a single attribute, then it is called a simple Candidate key. { eg: ck = (Sid) in eg. (1) }

(2) If Candidate key is formed using two or more attributes, then it is called Compound or Composite Candidate key

(3) A relation may have more than one Candidate keys.



Topic : Candidate key

④ Attributes belonging to any of the Candidate key of the relation are called prime attributes of the relation., all other attributes are called Non-prime attribute

Prime attributes ≡ key attributes

Non-prime attributes ≡ Non-key attributes

⑤ Candidate key attributes may take null values



Topic : Primary key

* There may be more than one candidate key in a relation. One of the candidate key is choosen as primary key.

* Primary key attributes are not allowed to accept the null values.

* Primary key values are unique and not NULL.



Topic : Alternate key

* Candidate keys except the primary key are called alternate keys.

Note: Candidate key, Primary key and alternate key, all are minimal keys.

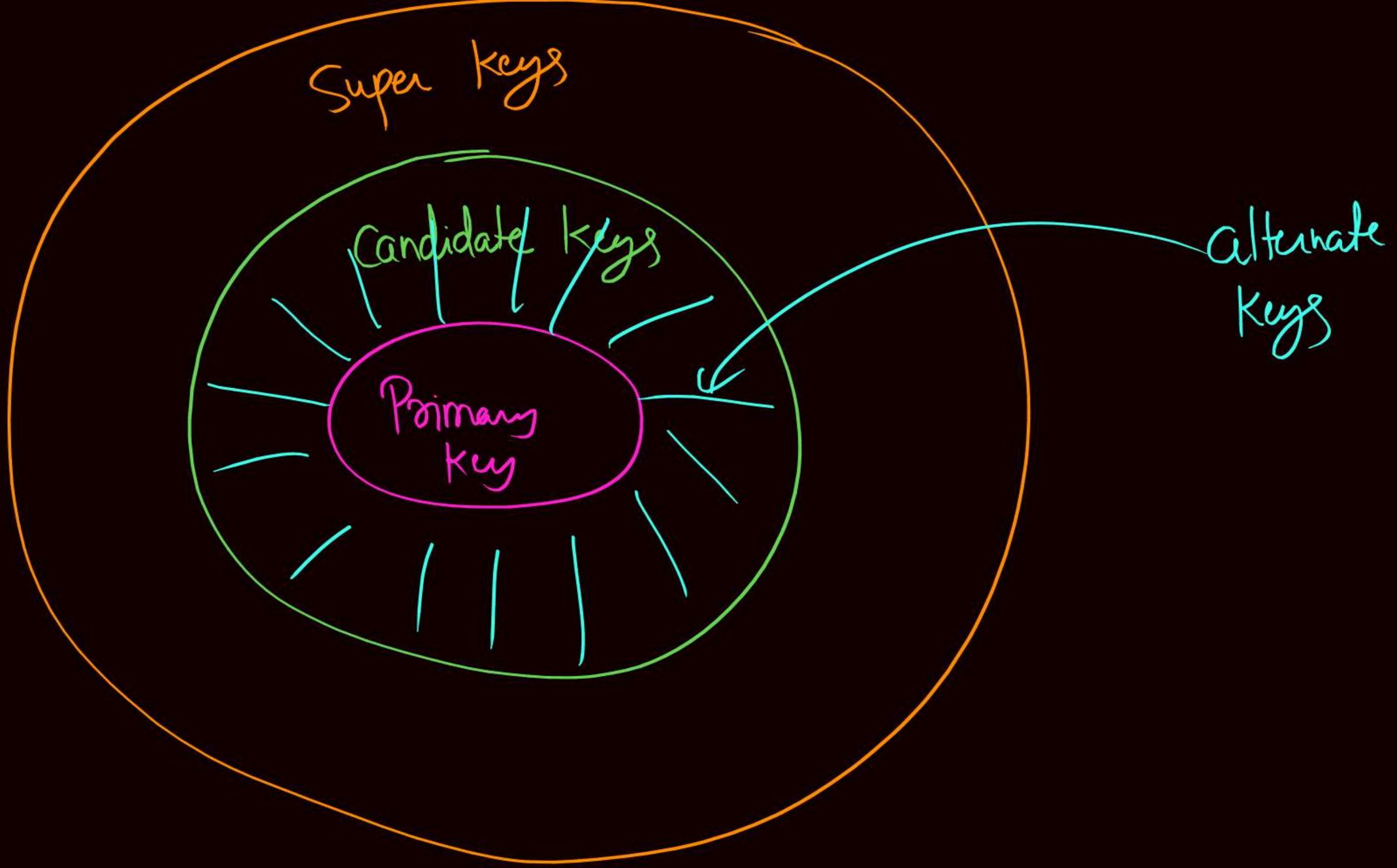


Topic : Super key

→ Any set of attributes {not necessarily minimal} that can differentiate each record of the table uniquely is called a Super key { Need not be a minimal set }

{ It may be minimal
or
may not be minimal }

i.e. Every candidate key is a Super key, but every Super key need not be a Candidate key


Super keys
Candidate keys
Primary key
alternate keys

eg ① Consider the Following Relational schema

Student (Sid, Sname, Branch)

let following FDs exists in the Relⁿ

Sid → Sname

Sid → Branch

Sid	Sname	Branch
S_1	A	CS
S_2	A	CS
S_3	B	CS
S_4	C	IT
S_5	B	IT

In eg ① Sid is the Only Candidate key.

Q: Find all the Superkeys of the relation student

If Sid is unique, then any superset of Sid will also have unique value.

∴ Every superset of Sid is a Superkey

∴ Superkeys of Relⁿ are

① (Sid)

② (Sid, Sname)

③ (Sid, Branch)

④ (Sid, Sname, Branch)



Topic : Super key

Note:

1. At least one subset of Super key is a Candidate key.
2. Super key = Attributes of any one of the Candidate key + 0 or More attributes from Remaining attributes of Relation



2 mins Summary

- **Topic** Different types of keys in RDBMS
- **Topic** Closure of an attribute set
- **Topic** Super key
- **Topic** Candidate key

THANK - YOU